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Note: during the solving of this assignment i have some doubt in quotations so i can solve this those questions in both may and mention that pdf
I write because of an early assignment. I also mentioned these things but sir did not read those details and gave negative feedback to me. It's good for me to improve but I mention the things I am facing, that time issue and reason also then resolve this issue myself.

A.

```
SELECT name,description,price,category
FROM Products p
WHERE price > (
    SELECT AVG(price)
    FROM Products
    WHERE category = p.category
)
ORDER BY category ASC, price DESC;
```

name	description	price	category
Watch	Analog Watch	90.00	Accessories
Book	Programming Book	30.00	Books
Jacket	Winter Jacket	80.00	Clothing
Shoes	Sports Shoes	60.00	Clothing
Laptop	Gaming Laptop	800.00	Electronics
Camera	DSLR Camera	500.00	Electronics
Smartphone	Android Phone	300.00	Electronics
Tablet	Android Tablet	250.00	Electronics

B. In this question we solve two ways because I have doubts about whether furniture is considered or not in ans.

one way there is consider "furniture" also in ans

```
SELECT *
FROM Products
WHERE price > ANY (
    SELECT price FROM Products WHERE category = 'Furniture'
);
```

product_id	name	description	price	category
1	Laptop	Gaming Laptop	800.00	Electronics
2	Smartphone	Android Phone	300.00	Electronics
6	Monitor	24 inch Monitor	200.00	Electronics
7	Tablet	Android Tablet	250.00	Electronics
9	Camera	DSLR Camera	500.00	Electronics
21	Sofa Set	Comfortable 3-seater sofa	600.00	Furniture
22	Dining Table	Wooden dining table	450.00	Furniture
24	Bed	Queen size wooden bed	700.00	Furniture
25	Wardrobe	3-door wardrobe	550.00	Furniture
26	Coffee Table	Modern coffee table	150.00	Furniture
27	Bookshelf	5-layer bookshelf	200.00	Furniture
28	TV Unit	TV stand cabinet	300.00	Furniture
NULL	NULL	NULL	NULL	NULL

“furniture” is not considered in ans

```
SELECT product_id , name , description , price , category
FROM Products
WHERE category != 'Furniture'
AND price > ANY (
    SELECT price
    FROM Products
    WHERE category = 'Furniture'
);
```

product_id	name	description	price	category
1	Laptop	Gaming Laptop	800.00	Electronics
2	Smartphone	Android Phone	300.00	Electronics
6	Monitor	24 inch Monitor	200.00	Electronics
7	Tablet	Android Tablet	250.00	Electronics
9	Camera	DSLR Camera	500.00	Electronics
NULL	NULL	NULL	NULL	NULL

C.

```
DELIMITER $$
CREATE PROCEDURE ApplyDiscount(IN target_cat VARCHAR(50), IN
discount_percent INT)
BEGIN
    UPDATE Products
    SET price = price - (price * discount_percent / 100)
```

```
    WHERE category = target_cat;
END $$
DELIMITER ;
```

```
13  19:20:10  CREATE PROCEDURE ApplyDiscount(IN target_cat VARCHAR(50), IN discount_percent INT) BEGIN  ...  0 row(s) affected
```

D. For execute this query first need to add one more colum

```
14  19:39:19  ALTER TABLE Customers ADD total_spent DECIMAL(10,2) DEFAULT 0  0 row(s) affected Records: 0  Duplicates: 0  Warnings: 0
```

Then create the procedure

```
DELIMITER $$
CREATE TRIGGER update_spending
AFTER INSERT ON Orders
FOR EACH ROW
BEGIN
    UPDATE Customers
    SET total_spent = total_spent + NEW.total_amount
    WHERE customer_id = NEW.customer_id;
END $$
DELIMITER ;
```

```
15  19:42:18  CREATE TRIGGER update_spending AFTER INSERT ON Orders FOR EACH ROW BEGIN  UPDATE...  0 row(s) affected
```

E.

```
SELECT
    c.customer_id,
    c.name,
    COALESCE(o.order_id, 'No orders placed') AS order_info
FROM Customers c
LEFT JOIN Orders o
ON c.customer_id = o.customer_id
AND o.order_date = (
    SELECT MAX(order_date)
    FROM Orders o2
    WHERE o2.customer_id = c.customer_id
);
```

customer_id	name	order_info
1	Ravi Kumar	1
2	Aman Singh	2
3	Priya Sharma	3
4	Neha Verma	4
5	Rahul Das	5
6	Sneha Gupta	6
7	Vikas Yadav	7
8	Pooja Mishra	8
9	Ankit Jain	9
10	Karan Patel	10
11	Rohit Mehta	11
12	Simran Kaur	12
13	Deepak Sha...	13
14	Nikita Roy	14
15	Suresh Reddy	15
16	Meena Joshi	16
17	Ajay Thakur	17
18	Divya Kapoor	18
19	Arjun Nair	19
20	Kavita Iyer	20
21	Test User	No orders...

F.

```
SELECT
   CASE
       WHEN email = mobile THEN NULL
       ELSE email
   END AS email,
   mobile , name
FROM Customers;
```

email	mobile	name
neha4@gmail.com	9000000004	Neha Verma
rahul5@gmail.com	9000000005	Rahul Das
sneha6@gmail.com	9000000006	Sneha Gupta
vikas7@gmail.com	9000000007	Vikas Yadav
pooja8@gmail.com	9000000008	Pooja Mishra
ankit9@gmail.com	9000000009	Ankit Jain
karan10@gmail.com	9000000010	Karan Patel
rohit11@gmail.com	9000000011	Rohit Mehta
simran12@gmail.com	9000000012	Simran Kaur
deepak13@gmail.com	9000000013	Deepak Sha...
nikita14@gmail.com	9000000014	Nikita Roy
suresh15@gmail.com	9000000015	Suresh Reddy
meena16@gmail.com	9000000016	Meena Joshi
ajay17@gmail.com	9000000017	Ajay Thakur
divya18@gmail.com	9000000018	Divya Kapoor
arjun19@gmail.com	9000000019	Arjun Nair
kavita20@gmail.com	9000000020	Kavita Iyer
testuser@gmail.com	9999999999	Test User
NULL	ravi@gmail....	Ravi
aman@gmail.com	9876543210	Aman
NULL	neha@yah...	Neha
priya@gmail.com	9123456780	Priya
NULL	karan@gma...	Karan

G.

```
 SELECT order_id, total_amount,
CASE
    WHEN total_amount < 100 THEN 'Low Value'
    WHEN total_amount BETWEEN 100 AND 500 THEN 'Medium Value'
    ELSE 'High Value'
END AS category
FROM Orders;
```

order_id	total_amount	category
1	800.00	High Value
2	600.00	High Value
3	100.00	Medium Value
4	210.00	Medium Value
5	80.00	Low Value
6	200.00	Medium Value
7	500.00	Medium Value
8	100.00	Medium Value
9	500.00	Medium Value
10	240.00	Medium Value
11	60.00	Low Value
12	80.00	Low Value
13	80.00	Low Value
14	120.00	Medium Value
15	90.00	Low Value
16	140.00	Medium Value
17	50.00	Low Value
18	90.00	Low Value
19	50.00	Low Value
20	50.00	Low Value

H.

```
SELECT MONTH(order_date) AS month, COUNT(*) AS total_orders
FROM Orders
GROUP BY MONTH(order_date);
```

month	total_orders
3	24
1	4
2	4
4	4
5	4

I.

```
SELECT product_id, name, NULLIF(discount, 0) AS discount
FROM Products;
```

product_id	name	discount
1	Laptop	10
2	Smartphone	NULL
3	Headphones	5
4	Keyboard	NULL
5	Mouse	15
6	Monitor	NULL
7	Tablet	20
8	Charger	NULL
9	Camera	25
10	Speaker	NULL
11	T-shirt	NULL
12	Jeans	NULL
13	Jacket	NULL
14	Shoes	NULL
15	Watch	NULL
16	Bag	NULL
17	Perfume	NULL
18	Book	NULL
19	Notebook	NULL

J.

```
SELECT  order_id, quantity ,total_amount , total_amount / NULLIF(quantity, 0) AS value_per_item
FROM Orders;
```

order_id	quantity	total_amount	value_per_item
27	2	500.00	250.000000
28	4	100.00	25.000000
29	1	500.00	500.000000
30	2	240.00	120.000000
31	3	60.00	20.000000
32	2	80.00	40.000000
33	1	80.00	80.000000
34	2	120.00	60.000000
35	1	90.00	90.000000
36	2	140.00	70.000000
37	1	50.00	50.000000
38	3	90.00	30.000000
39	5	50.00	10.000000
40	10	50.00	5.000000
41	0	0.00	NULL
42	0	0.00	NULL
43	0	0.00	NULL
44	0	0.00	NULL
45	0	0.00	NULL

K.

For solving this query i have no any column that have the score column so i can create another table for that and insert some data

I also user this operation in another table but according to assignment i can create this table because prev i change something on my then give negative feedback

```
CREATE TABLE Students (
   id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(50),
   score INT
);

INSERT INTO Students (name, score) VALUES
('Ravi', 10),
('Aman', 0),
('Neha', 25),
('Priya', 0),
('Karan', 15),
('Ankit', 0),
('Pooja', 30);

SELECT COUNT(*) AS non_zero_count
FROM Students
```

```
WHERE score <> 0;
```

	non_zero_count
▶	4

L.

```
CREATE TABLE Employees (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    salary INT
);

INSERT INTO Employees (name, salary) VALUES
('Ravi', 50000),
('Aman', 0),
('Neha', 60000),
('Priya', 0),
('Karan', 45000),
('Ankit', 70000);
– Solve in two way give some output
SELECT AVG(salary) AS avg_salary
FROM Employees
WHERE salary <> 0;

SELECT AVG(NULLIF(salary, 0)) AS avg_salary
FROM Employees;
```

	avg_salary
▶	56250.0000

M.

Use to order table to solve this query

```
SELECT
    order_id,
    total_amount,
    (total_amount / NULLIF((SELECT SUM(total_amount) FROM Orders), 0)) * 100
AS percentage
FROM Orders;
```

order_id	total_amount	percentage
19	50.00	0.603900
20	50.00	0.603900
21	800.00	9.661800
22	600.00	7.246400
23	100.00	1.207700
24	210.00	2.536200
25	80.00	0.966200
26	200.00	2.415500
27	500.00	6.038600
28	100.00	1.207700
29	500.00	6.038600
30	240.00	2.898600
31	60.00	0.724600
32	80.00	0.966200
33	80.00	0.966200
34	120.00	1.449300
35	90.00	1.087000
36	140.00	1.690800
37	50.00	0.603900
38	90.00	1.087000
39	50.00	0.603900
40	50.00	0.603900
41	0.00	0.000000
42	0.00	0.000000
43	0.00	0.000000

N.

```
SELECT
    NULLIF(email, '') AS email
FROM Customers;
```

customer_id	email
30	NULL
17	ajay17@gmail.com
23	aman@gmail.com
2	aman2@gmail.com
9	ankit9@gmail.com
19	arjun19@gmail.com
13	deepak13@gmail.com
18	divya18@gmail.com
26	karan@gmail.com
10	karan10@gmail.com
20	kavita20@gmail.com
16	meena16@gmail.com
24	neha@yahoo.com
4	neha4@gmail.com
14	nikita14@gmail.com

O.

```
CREATE TABLE Projects (
   id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(50),
   start_date DATE,
   end_date DATE
);

INSERT INTO Projects (name, start_date, end_date) VALUES
('Project A', '2025-01-01', '2025-01-01'),  -- equal
('Project B', '2025-02-01', '2025-03-01'),  -- different
('Project C', '2025-04-10', '2025-04-10'),  -- equal
('Project D', '2025-05-01', '2025-06-01');  -- different


SELECT
   name,
   NULLIF(start_date, end_date) AS result
FROM Projects;
```

name	result
Project A	NULL
Project B	2025-02-01
Project C	NULL
Project D	2025-05-01

P.

```
SELECT AVG(NULLIF(total_amount, -1)) AS avg_spending
FROM Orders;
```

avg_spending
184.000000

;

Q.

```
SELECT
    name,
    CASE
        WHEN NULLIF(email, mobile) IS NULL THEN NULL
        ELSE email
    END AS cleaned_email
FROM Customers;
```

name	cleaned_email
Sneha Gupta	sneha6@gmail.com
Vikas Yadav	vikas7@gmail.com
Pooja Mishra	pooja8@gmail.com
Ankit Jain	ankit9@gmail.com
Karan Patel	karan10@gmail.com
Rohit Mehta	rohit11@gmail.com
Simran Kaur	simran12@gmail.com
Deepak Sha...	deepak13@gmail.com
Nikita Roy	nikita14@gmail.com
Suresh Reddy	suresh15@gmail.com
Meena Joshi	meena16@gmail.com
Ajay Thakur	ajay17@gmail.com
Divya Kapoor	divya18@gmail.com
Arjun Nair	arjun19@gmail.com
Kavita Iyer	kavita20@gmail.com
Test User	testuser@gmail.com
Ravi	NULL
Aman	aman@gmail.com
Neha	NULL
Priya	priya@gmail.com
Karan	NULL
Test1	
Test2	user@gmail.com
Test3	sdfsdf